

Summary

Innovative Software Engineer and Researcher with a proven track record of deploying scalable AI-driven solutions and robust software infrastructure. Expert in bridging the gap between cutting-edge AI research and production-ready applications, demonstrated by implementing RAG-based generative AI systems that increased code explanation accuracy from 30% to 90%. Experienced in building high-impact developer tools and E2E testing frameworks used by thousands of users across global enterprises like Snowflake and IBM. Highly proficient in Python, TypeScript, and cloud environments (AWS), with specialized research experience in AI alignment. Passionate about leveraging frontier AI technologies to solve complex problems and drive business impact through rapid prototyping and scalable engineering.

Work Experience

- Ramp** Waterloo, Canada
Incoming Software Engineer Intern Sept. 2025 - Dec. 2025
 - Selected for upcoming software engineering internship to contribute to high-scale financial technology solutions.
- University of Toronto** Waterloo, Canada
Research Intern Now - Now
 - Contribute to a university-partnered study with OpenAI focusing on AI alignment in the workplace.
 - Conduct advanced research under the supervision of Dr. Meena Andiappan and Dr. Gillian Hadfield to ensure safe and beneficial AI deployment.
- Snowflake** Waterloo, Canada
Software Engineer (Contract) Now - Now
 - Built AI Copilot for Notebooks, accelerating workflows for over 1,000 data scientists across 300+ organizations.
 - Owned the end-to-end development of resource monitoring in Notebooks using Python, gRPC, Protocol Buffers, React.js, and TypeScript.
 - Provided customers with real-time visibility into CPU, disk, and GPU usage via Vega Lite visualizations.
- Snowflake** Waterloo, Canada
Software Engineer Intern Now - Now
 - Designed and implemented an E2E testing framework adopted by 100+ engineers across 15+ teams using Jenkins, Docker, Groovy, and Datadog.
 - Architected UI component APIs consumed by developers on 5+ teams using React.js and TypeScript.
 - Deployed user experience enhancements across products utilized by 15,000+ users across 3,000+ organizations.
- TD Bank** Waterloo, Canada
Software Developer Intern Now - Now

- Designed a distributed task queue to asynchronously execute portfolio model calculations using Celery and Python.
- Successfully reduced manual workload for Portfolio Managers by saving 10+ hours per week previously lost to synchronous execution waiting.

- **IBM** Waterloo, Canada
Software Developer Intern *Now - Now*
 - Shipped a generative AI application for code explanation and test-case generation on 12M+ lines of code using AWS and IBM's WatsonX AI platform.
 - Increased code explanation accuracy from 30% to 90% by implementing Retrieval-Augmented Generation (RAG) with LangChain, FAISS vector database, and the Mixtral 8x7B LLM.
 - Engineered a rules-based anomaly detection system to verify \$10B+ in insurance claims using Python and SQL.

Projects

- **Pethsapp** *Jun. 2021 - Jun. 2023*
Team Lead / Developer
 - Led a team of 5 to develop a mobile application for a high school community, serving 1200+ students, staff, and parents.
 - Achieved a Top 15 ranking on the Apple App Store Education Chart.

Education

- **University of Waterloo** Waterloo, Canada
Bachelor of Computer Science *2022 - 2026*

Courses & Certifications

- **Operating Systems** University of Waterloo
Now
- **Statistics** University of Waterloo
Now
- **Object-Oriented Programming** University of Waterloo
Now
- **Probability** University of Waterloo
Now
- **Algorithms** University of Waterloo
Now
- **Data Structures** University of Waterloo
Now

Technical Skills

Python, TypeScript, JavaScript, C++, C, SQL, PostgreSQL, SQL Server, React.js, Next.js, Node.js, React Native, Tailwind CSS, CSS, HTML, StyleX, LangChain, FAISS, Mixtral 8x7B LLM, RAG (Retrieval-Augmented Generation), AWS, Google Cloud, Snowflake, Datadog, Docker, Jenkins, Groovy, Bash, gRPC, Protocol Buffers, Celery, Redis, MongoDB, Flask, Django, Vega Lite